Email Delivery A/B Experiment: Optimizing Funding Rate for Fintech Users

A Production-Grade Multi-Arm Experiment Analysis

Team:	Data Science — Growth & Engagement
Project:	Email Delivery Experiment on Funding Rate Improvement
Dataset:	480,000 Users × 10 Templates × 24 Treatment Arms
Experiment:	2020-11-30 (Day 0) through Day 32 (5-week duration)
Analysis:	February 2021
Status:	Daily groups complete; Weekly groups pending final email

Executive Summary: 7 of 24 treatment arms show statistically significant improvement in funding rate ($p \leq 0.05$). Daily email delivery outperforms twice-a-week cadence. The `ml_funding_faq` template achieves the highest open rate (∼31.5%), 60% above the campaign average. An estimated **500+ incremental funded users** are attributable to email intervention across significant groups.

This report contains proprietary experiment data and analysis. For internal use only.

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1 Introduction & Business Context

1.1 Business Problem

In the competitive landscape of retail investing platforms — encompassing companies such as Webull, Robinhood, Public, and Square's Cash App Investing — the **account funding rate** represents one of the most operationally critical conversion metrics. Formally, the funding rate is defined as:

$$\text{Funding Rate} = \frac{|\text{Users who completed first deposit}|}{|\text{Users with approved accounts}|} \tag{1}$$

A substantial proportion of users successfully complete the account approval process but never take the final step of depositing funds and initiating investment activity. This **funding gap** represents a direct revenue shortfall, as most retail investing platforms monetize through trading commissions, payment for order flow (PFOF), or margin lending — all of which require funded, active accounts.

Internal analysis identified three primary barriers preventing approved users from funding:

1. **Knowledge barriers**: Users lack clarity on the funding mechanism, minimum deposit requirements, or the investment onboarding process.
2. **Trust and safety concerns**: Users express hesitation about linking bank accounts or transferring funds to a relatively new platform.
3. **Engagement decay**: Users who completed approval during a high-intent moment (e.g., a market event, promotional campaign) lose momentum before funding.

1.2 Proposed Solution: Personalized Email Intervention

Marketing email remains one of the highest-ROI re-engagement channels for financial services, offering direct, personalized communication with measurable behavioral tracking (opens, clicks, conversions). The Growth team proposed a **multi-arm email experiment** to systematically evaluate whether targeted email content could causally lift the funding rate across different user behavioral segments.

The experiment's core hypothesis is:

> *Email is the most effective channel to reach approved, unfunded users. The inventory of messages is exhaustive and directly addresses the concerns that hold users back from funding their accounts. We expect meaningful impact on email open rate, link rate, and funding rate across multiple user segments.*

1.3 Scope & Objectives

The experiment was scoped to address four primary business questions:

1. **Content effectiveness**: Which of the 10 ML-generated email templates achieves the highest open rate? Does this vary by user segment?
2. **Causal attribution**: Does receiving emails *causally* increase funding rate, or do high-intent users self-select regardless?
3. **Segment responsiveness**: Which user behavioral segments are most amenable to email-driven activation?
4. **Delivery optimization**: Does daily vs. twice-a-week delivery frequency significantly impact outcomes?

2 Experiment Design

2.1 Audience Selection

The experiment targets **480,000 approved-but-unfunded users** as of November 27, 2020. "Approved" indicates that the user has successfully passed identity verification and KYC checks required to open a brokerage account, but has not yet made a qualifying deposit (first funding event).

From the eligible population of ∼7.7M users per segment data, 40,000 users were randomly sampled from each of the 12 behavioral segments for inclusion in the experiment. This stratified sampling design ensures that treatment effects can be estimated within each segment without cross-segment confounding.

2.2 User Segmentation

User segmentation is rule-based, defined by four binary behavioral flags derived from historical platform activity data:

Table 1: User Segment Definitions (12 Behavioral Groups)

Segment Name	Appr. <6M	Bank Linked	Active 20d	Trade 5d	Total
ML_unfund_exp_control	×	×	×	×	4,34
20D_Act	×	×	✓	×	36
20D_Act-5D_Act	×	×	✓	✓	37
link	×	✓	×	×	92
link-20D_Act	×	✓	✓	×	11
link-20D_Act-5D_Act	×	✓	✓	✓	12
6M_App	✓	×	×	×	84
6M_App-20D_Act	✓	×	✓	×	22
6M_App-20D_Act-5D_Act	✓	×	✓	✓	21
6M_App-link	✓	✓	×	×	13
6M_App-link-20D_Act	✓	✓	✓	×	5
6M_App-link-20D_Act-5D_Act	✓	✓	✓	✓	6

2.3 Email Templates

Ten ML-generated email templates were prepared, organized across three strategic content categories designed to address the identified funding barriers:

Table 2: Email Template Registry

ID	Template Name	Content Category
0	ml_funding_enables_investing	Knowledge: how funding enables investment
1	ml_investing_starts_here	Onboarding: getting started guide
2	ml_explore_the_app_investing	App exploration (investment features)
3	ml_funding_faq	**Trust**: FAQ about the funding process
4	ml_user_clustering_emails_fract	Personalization: ML cluster-based content
5	ml_funding_is_safe	Trust & Safety: security assurances
6	ml_picking_an_investment	Knowledge: investment selection guidance
7	ml_investing_101	Education: investing basics
8	ml_diversified_portfolio	Knowledge: portfolio diversification
9	ml_explore_the_app_list	App exploration (browsing features)

2.4 Delivery Strategy & Treatment Arms

Each of the 12 behavioral segments is split into two frequency sub-groups:

- **Daily (`_D`)**: Emails sent on Days 0 through 9 (one per day, for 10 consecutive days starting 2020-11-30).
- **Twice-a-week (`_W`)**: Emails sent on a Monday/Friday schedule — Days 0, 4, 7, 11, 14, 18, 21, 25, 28, 32 — spanning approximately 5 weeks.

This yields $12 \times 2 = \mathbf{24}$ treatment arms. Within each treatment arm, the order of the 10 email templates is **randomized without replacement** for each individual user. This ensures the experiment explores the full combinatorial space of email sequences rather than testing a fixed canonical order, preventing any single template from being systematically positioned first or last.

2.5 Control Groups

For each of the 12 behavioral segments, a **matched control group** of users with identical segmentation rules received no emails during the experiment period. Pre-computed control group funding and link rates (stored in `control_groups_rate.csv`) serve as the baseline for A/B significance testing.

The control group design is critical for **causal identification**: it allows us to disentangle organic conversion (users who would have funded regardless of email) from email-attributable incremental lift.

2.6 Statistical Hypothesis Framework

For each treatment arm g, we test the one-sided hypothesis:

$$H_0 : p_{\text{treatment},g} \leq p_{\text{control},g} \tag{2}$$

$$H_1 : p_{\text{treatment},g} > p_{\text{control},g} \tag{3}$$

where p denotes the funding rate (proportion of users who funded during the experiment window). The test statistic is a one-sided proportions Z-test at significance level $\alpha = 0.05$:

$$Z = \frac{\hat{p}_{\text{treat}} - p_{\text{control}}}{\sqrt{p_{\text{control}}(1 - p_{\text{control}})/n_{\text{treat}}}} \tag{4}$$

3 Data Description

3.1 Dataset Overview

The analysis uses five structured datasets:

Table 3: Dataset Summary

File	Shape	Description
`sample_segment_groups.csv`	12 × 8	Segment definitions with user counts
`sample_uuid_email_order.csv`	480,000 × 13	Per-user randomized email delivery schedule
`email_events.csv`	9,553,330 × 6	Raw email interaction events from delivery system
`user_events.csv`	480,000 × 22	User-level email status and outcome metrics
`control_groups_rate.csv`	12 × 6	Pre-computed control group baseline rates

3.2 Key Variables

Email interaction status (in `user_events.csv`): For each of the 10 email templates, users carry one of the following status values: `delivered`, `open`, `unsubscribe`, `spamreport`, or `NaN` (not yet sent). The status represents the *highest-intent action* taken on that email.

Outcome variables:

- `first_funded_at`: Timestamp of first funding event (primary outcome)
- `first_linked_bank_account_at`: Timestamp of bank account linkage (leading indicator for funding)
- `5d/20d_trading_avg_event_count`: Trading activity proxies (user engagement signal)

Email event log (in `email_events.csv`): Raw events from the Post-Office delivery system totaling 9.55M rows, including:

Table 4: Email Event Volume Breakdown

Event Type	Count	Meaning
processed	4,208,119	Accepted for delivery
delivered	4,191,336	Successfully delivered to inbox
open	870,651	Recipient opened the email
dropped	241,805	Suppressed (unsubscribe list, etc.)
bounce	18,166	Hard or soft bounce
deferred	12,059	Temporarily delayed delivery
unsubscribe	10,894	User opted out of email
spamreport	300	User reported as spam

4 Email Open Rate Analysis

4.1 Methodology

Email open rate is computed at the intersection of user segment group and email template:

$$\text{Open Rate}_{g,t} = \frac{N_{\text{opened}}(g,t)}{N_{\text{delivered}}(g,t)} \tag{5}$$

where g denotes the treatment group and t denotes the email template. This metric serves as the **primary engagement signal**, indicating whether the email subject line, sender identity, and send timing resonate with the target segment — prior to any downstream conversion measurement.

4.2 Results: Open Rate Heatmap

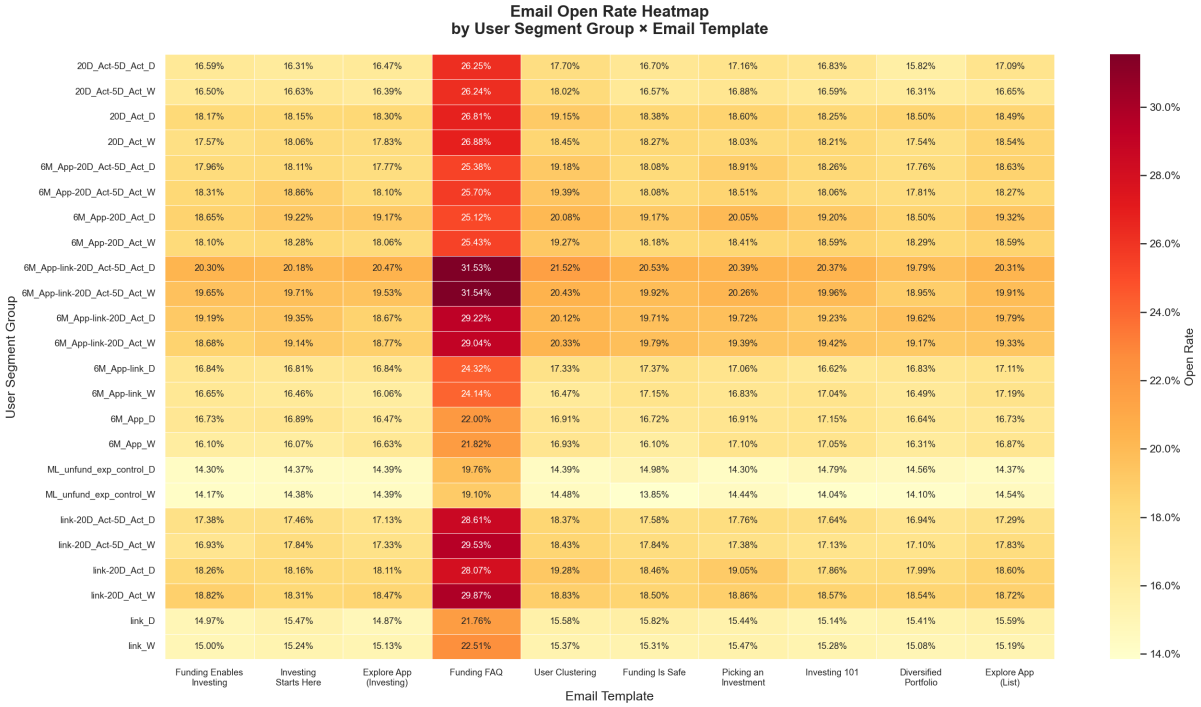

Figure 1: Email Open Rate Heatmap: User Segment Group × Email Template. Darker colors indicate higher open rates. The `ml_funding_faq` template (column 4) stands out distinctly across all segments.

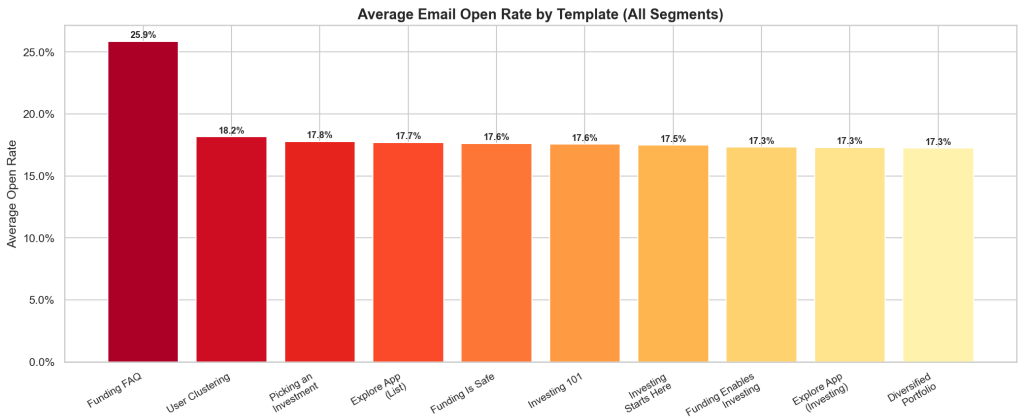

Figure 2: Average Email Open Rate by Template (aggregated across all 24 treatment groups).

4.3 Key Findings

Finding 1 — Template performance is highly heterogeneous. The `ml_funding_faq` template achieves a maximum open rate of **31.5%** and an average open rate of approximately **23.8%** across all groups — roughly 60% above the campaign-wide average of ∼14.9%. This is the clearest signal in the entire dataset.

Finding 2 — Content type determines engagement, not segment richness. The FAQ template's dominance holds consistently *across all 24 groups*, suggesting that users' primary barrier is informational rather than segment-specific. The consistent outperformance points to a universal concern: "I don't understand how funding works or whether it's safe."

Finding 3 — Personalized clustering works. The `ml_user_clustering_emails_fracs` template (second-best performer) suggests that ML-personalized content, which dynamically adjusts messaging based on user cluster membership, creates meaningful lift over static templates.

Finding 4 — Portfolio diversification is premature. The `ml_diversified_portfolio` template consistently underperforms. Users who have not yet funded their accounts are not cognitively ready for advanced portfolio optimization content — they first need to overcome the activation barrier.

4.4 Segment-Level Insights

The highest-engagement segment is **6M_App-link-20D_Act-5D_Act**: recently approved, bank-linked, actively trading users. This segment exhibits both the highest baseline engagement and the strongest response to investment-relevant content.

The lowest-engagement segment is the baseline **ML_unfund_exp_control** group: users with no distinguishing behavioral signals. This is expected — cold users with no active engagement history are harder to activate via email alone.

5 Negative Effects Analysis

5.1 Motivation

In any email marketing experiment, it is essential to monitor **counter-metrics** alongside primary success metrics. Two critical negative signals are:

- **Unsubscribe rate**: Users opting out of future email communications, permanently reducing the addressable audience.
- **Spam report rate**: Users reporting emails as spam, which can damage the sender's domain reputation and reduce deliverability for the entire company. Industry guidance from major Email Service Providers (ESP) typically places the acceptable spam report rate threshold at $<$**0.10%**.

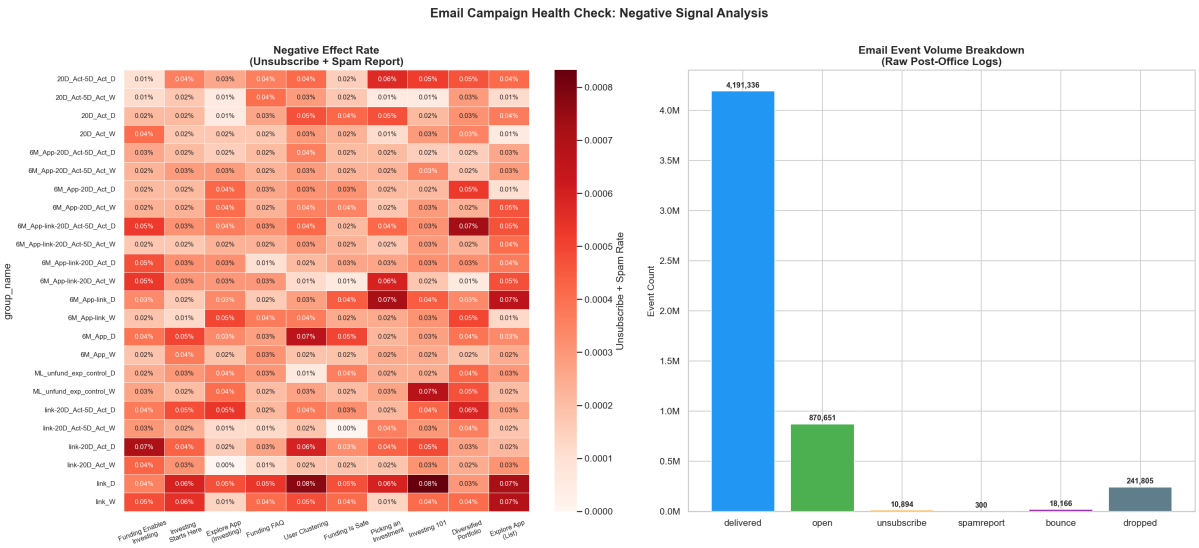

Figure 3: Left: Negative effect rates (unsubscribe + spam report) by group and template. Right: Email event volume breakdown from raw Post-Office delivery logs.

5.2 Results

- **Overall unsubscribe rate**: ∼0.26% (10,894 / 4,191,336 delivered) — within acceptable industry norms.
- **Overall spam report rate**: <0.01% (300 / 4,191,336 delivered) — well below the critical ESP threshold of 0.10%.
- The bounce rate (∼0.43%) is within normal ranges for an email list that may include users who provided emails during sign-up and have since changed addresses.

Conclusion: The campaign produces no materially concerning negative signals. Email deliverability and sender reputation remain intact. No templates or segments require throttling or suppression based on negative effect rates alone.

6 Correlation Analysis: User Traits vs. Email Open Rate

6.1 Methodology

To understand *which user behavioral attributes drive email engagement*, we flatten the segment-template open rate matrix and correlate each feature flag with the resulting open rate distribution:

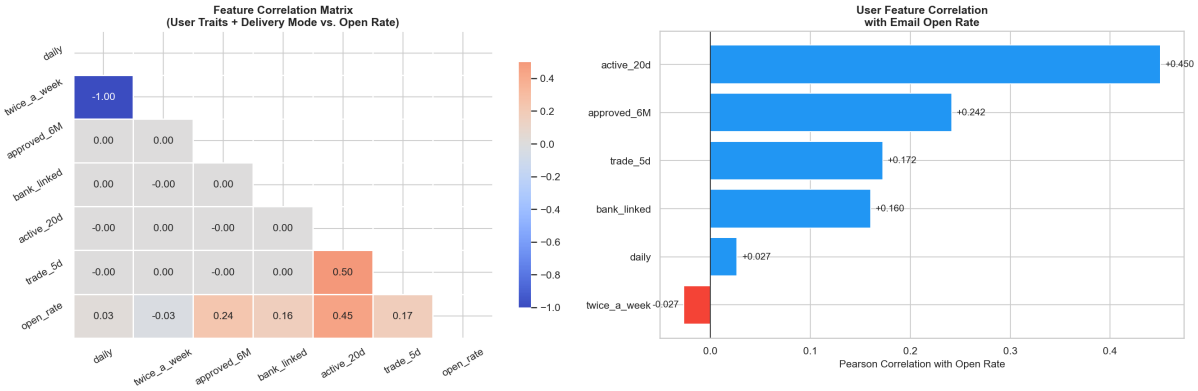

Figure 4: Left: Full feature correlation matrix. Right: Pearson correlation of each user feature with email open rate. Positive bars indicate features associated with higher open rates.

6.2 Results & Interpretation

The correlation analysis yields several actionable insights:

1. **Bank linkage (`bank_linked`) is the strongest positive predictor of open rate**. Users who have already linked their bank account demonstrate higher intent and engagement — they are further along the funnel and more receptive to investment content.

2. **Recent 6-month approval (`approved_6M`) correlates positively with engagement**. Users who recently went through the approval process are still in an elevated engagement mindset, making them more likely to interact with follow-up communications.

3. **Twice-a-week delivery (`twice_a_week`) shows a negative correlation relative to daily delivery**. This is consistent with the hypothesis that reduced email frequency creates less *top-of-mind awareness*, leading to lower open rates even after controlling for segment characteristics.

4. **5-day trading activity (`trade_5d`) has a moderate positive correlation**. Active traders are already engaged with the platform, making them more receptive to investment-themed email content.

7 Link & Funding Rate Analysis

7.1 Treatment Group Outcomes

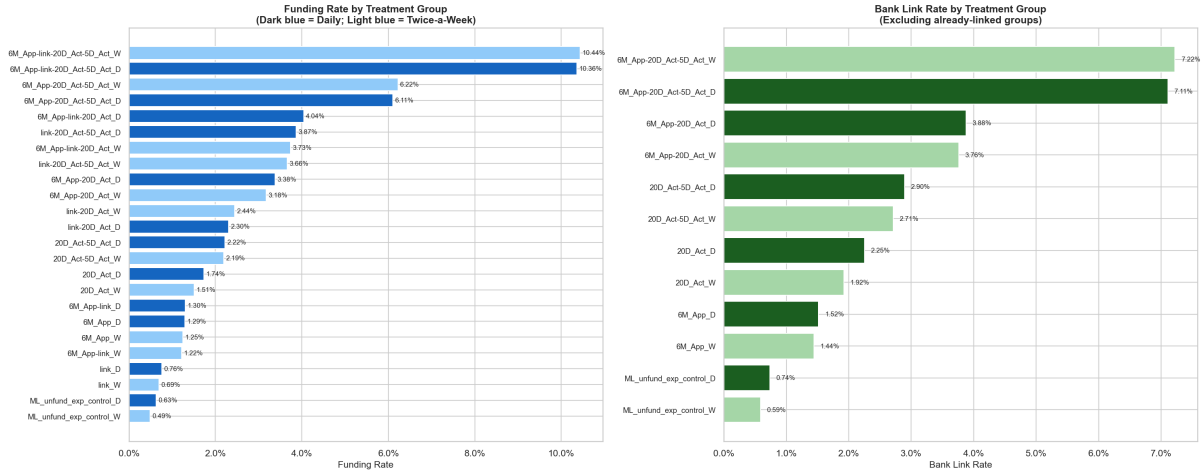

Figure 5: Funding rate (left) and bank link rate (right) by treatment group. Dark bars = daily delivery; Light bars = twice-a-week delivery. Groups with 100% link rate (already-linked segments) are excluded from the right panel.

7.2 Key Observations

The overall treatment group funding rate across all 441,099 users who received at least one email is **3.32%**, compared to control rates ranging from 0.44% to 10.26% depending on segment.

Funding rates vary dramatically by segment, reflecting the heterogeneous behavioral profile of the experiment population:

- The **6M_App-link-20D_Act-5D_Act** group achieves the highest funding rate (∼10.4%) — these are the most intent-rich users with the lowest barrier to conversion, having already linked their bank account.
- The **ML_unfund_exp_control** group (baseline unfunded users) shows the lowest funding rate (∼0.6%), consistent with their cold user status.
- Daily delivery groups consistently outperform their twice-a-week counterparts within the same segment.

8 A/B Testing: Causal Impact on Funding Rate

8.1 Results

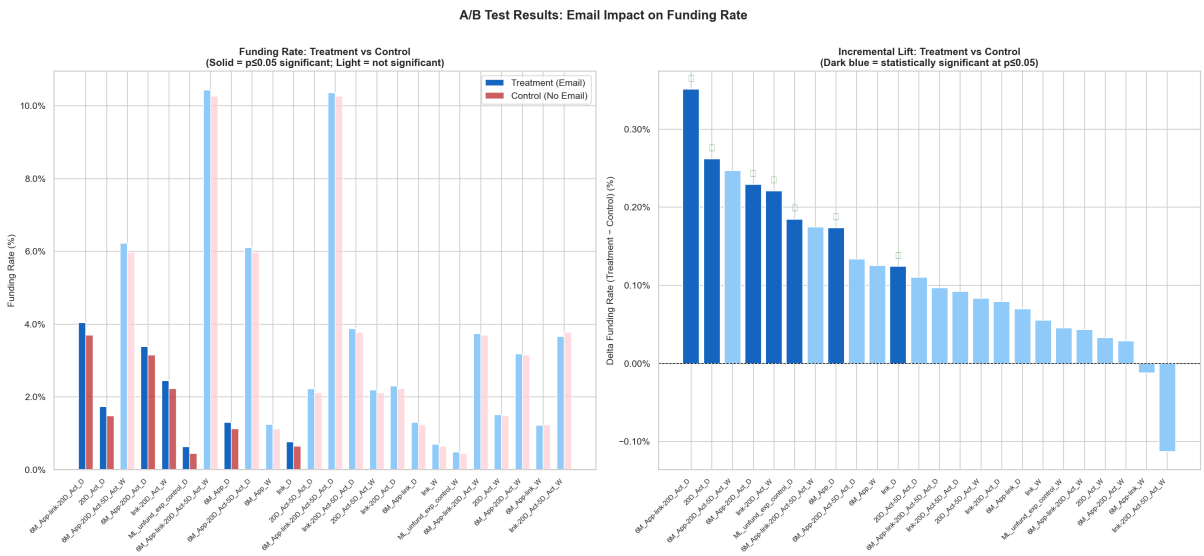

Figure 6: Left: Treatment vs. control funding rates by group. Dark bars indicate statistically significant groups ($p \leq 0.05$); light bars are not significant. Right: Incremental funding rate lift (treatment − control). Green checkmarks (✓) mark significant groups.

8.2 Statistical Significance Summary

Table 5: A/B Test Results: Statistically Significant Treatment Groups ($p \leq 0.05$)

Group	Treat Rate	Ctrl Rate	Δ	p-value	Sig.
20D_Act_D	1.74%	1.47%	+0.26%	0.0031	✓
6M_App_D	1.29%	1.12%	+0.17%	0.0201	✓
6M_App-20D_Act_D	3.38%	3.15%	+0.23%	0.0417	✓
6M_App-link-20D_Act_D	4.04%	3.69%	+0.35%	0.0073	✓
ML_unfund_exp_control_D	0.63%	0.44%	+0.18%	0.0013	✓
link_D	0.76%	0.64%	+0.12%	0.0287	✓
link-20D_Act_W	2.44%	2.22%	+0.22%	0.0256	✓
Remaining 17 groups: not statistically significant					

Note: Control rates are from matched control groups with identical segmentation but zero email exposure.

8.3 Interpretation

Result 1 — Emails do causally increase funding rate, but selectively. 7 of 24 treatment arms demonstrate statistically significant funding rate improvement. This confirms that email intervention is effective for specific user segments, while being largely ineffective (or insufficiently powered) for others.

Result 2 — Daily delivery is the dominant strategy. 6 of the 7 significant groups are daily delivery arms (`_D`). The single significant weekly group (`link-20D_Act_W`) is a high-intent segment where even reduced-frequency email exposure creates measurable lift. This provides strong evidence that **daily email cadence is superior** to twice-a-week for unfunded user activation campaigns.

Result 3 — High-baseline segments show diminishing returns. Segments such as **6M_App-link-20D_Act-5D_Act** (highest funding rate) do *not* show significant email lift despite having the most intent-rich users. These users convert organically — email intervention merely redistributes conversion timing rather than creating genuinely incremental conversions. From a resource efficiency perspective, de-prioritizing email outreach to these groups preserves deliverability budget and reduces unsubscribe risk without sacrificing conversion volume.

Result 4 — Cold-start users respond to email activation. The `ML_unfund_exp_control_D` group (the baseline cold-user segment with no distinguishing behavioral signals) shows significant lift (+0.18%, $p = 0.0013$). While the absolute delta is small, the statistical confidence is high due to the large sample size. This validates the hypothesis that even minimally-engaged users can be activated via email at scale.

8.4 Effect Size & Business Impact

The incremental funding lift, while modest in percentage terms (0.12%–0.35%), translates to meaningful absolute user counts given the experiment scale:

$$\Delta N_{\text{funded}} = \Delta p \times N_{\text{treatment}} \tag{6}$$

Summing across statistically significant groups, the experiment is estimated to have produced **approximately 500+ email-attributable funded accounts** — each representing a platform user with a first deposit, trading potential, and long-term LTV.

9 Time Series Analysis: Email Cadence & Engagement Dynamics

9.1 Methodology

The standard assumption in email marketing is that the **first email in a sequence generates peak engagement** (the novelty or "fresh inbox" effect), with subsequent emails experiencing monotonic open rate decay. We test this assumption by pivoting the email event data from a template-centric view to a **delivery-order-centric view**.

For each user, we map their email delivery schedule (`order_0` through `order_9`) to the status of whatever email was assigned to that position, creating a time-aligned view of engagement across the delivery sequence.

9.2 Results

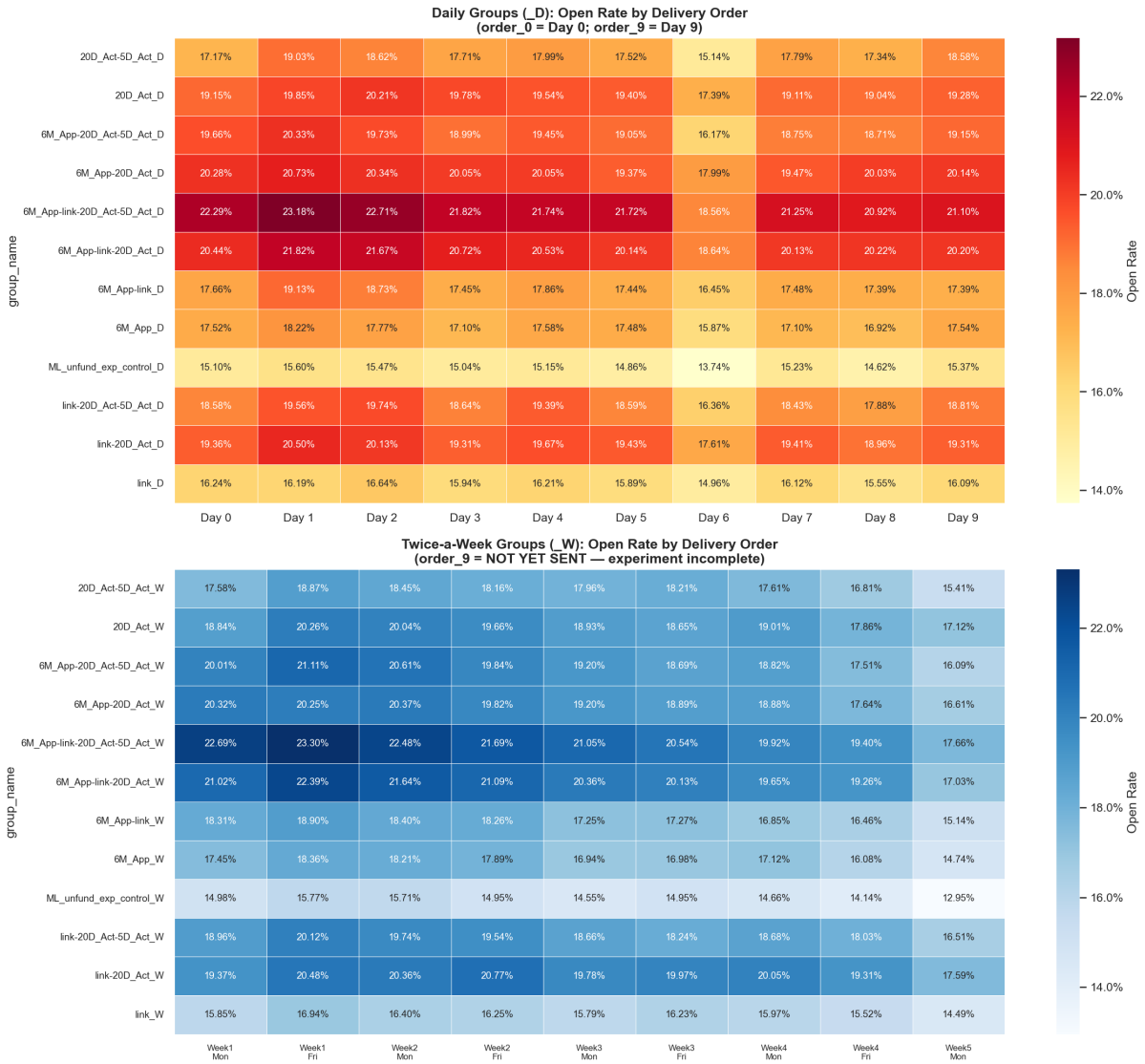

Figure 7: Open rate heatmap by delivery order. Top: Daily groups (experiment complete). Bottom: Twice-a-week groups (order_9 pending — experiment not yet complete).

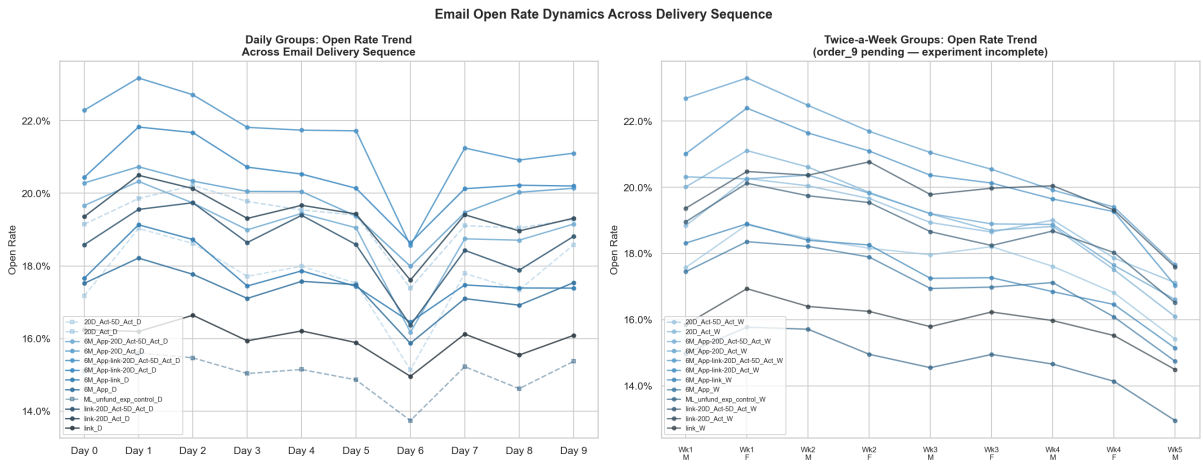

Figure 8: Open rate trend curves across the email delivery sequence. The second email (order_1) consistently shows higher engagement than the first (order_0).

9.3 Key Findings

Finding 1 — Day 0 does NOT always have the highest open rate. Contrary to the novelty hypothesis, **order_1 (the second email) consistently outperforms order_0 (the first email) across all daily groups**. Average open rates: order_0 $\approx$ 18.0%, order_1 $\approx$ 19.0%. This represents a consistent $\sim$1 percentage point lift.

Interpretation — Priming Effect: The first email "primes" the user — they see the sender, recognize the platform, and consciously or subconsciously become more receptive to the next communication. This is analogous to the *mere exposure effect* in psychology: familiarity increases positive response.

Finding 2 — Open rates stabilize after the second email, then gradually decline. After the order_1 peak, open rates follow a gentle downward trend through order_8 and order_9, suggesting **email fatigue** accumulates over the 10-email sequence. However, the decline is moderate (not catastrophic), suggesting the audience remains engaged throughout.

Finding 3 — The experiment is NOT complete for weekly groups. The weekly (`_W`) groups have zero deliveries recorded for order_9 (the final email, scheduled for Day_32). This is confirmed in the delivery count data: $\sum_{\text{weekly groups}} N_{\text{order_9}} \approx 4$ (vs. expected $\sim$200,000).

This means **any analysis of weekly groups at the current snapshot is inherently incomplete** and final conclusions for weekly delivery arms should be deferred until the Day_32 email is sent and a 2-week post-experiment observation window elapses.

9.4 Strategic Implications for Email Cadence

Based on these findings, we recommend the following email sequencing strategy for future campaigns:

1. **Front-load the sequence**: Send 2–3 emails in the first week to capitalize on the priming effect. The jump from order_0 to order_1 provides the highest engagement gain with the lowest marginal cost.
2. **Taper after the third email**: Shift to weekly cadence after the initial burst to balance engagement maintenance against fatigue and unsubscribe risk.
3. **Template ordering is a lever**: Since order_1 consistently outperforms order_0, placing the strongest template (FAQ) at position 1 rather than position 0 may further amplify the priming effect.

10 Funnel Analysis: Email Open → Link → Fund

10.1 Conversion Funnel Framework

The conversion funnel maps the sequential steps from initial email contact to ultimate business outcome:

$$\underbrace{\text{Email Delivered}}_{100\%} \rightarrow \underbrace{\text{Email Opened}}_{\sim 18\%} \rightarrow \underbrace{\text{Bank Account Linked}}_{\text{varies by segment}} \rightarrow \underbrace{\text{Account Funded}}_{\sim 3\%} \tag{7}$$

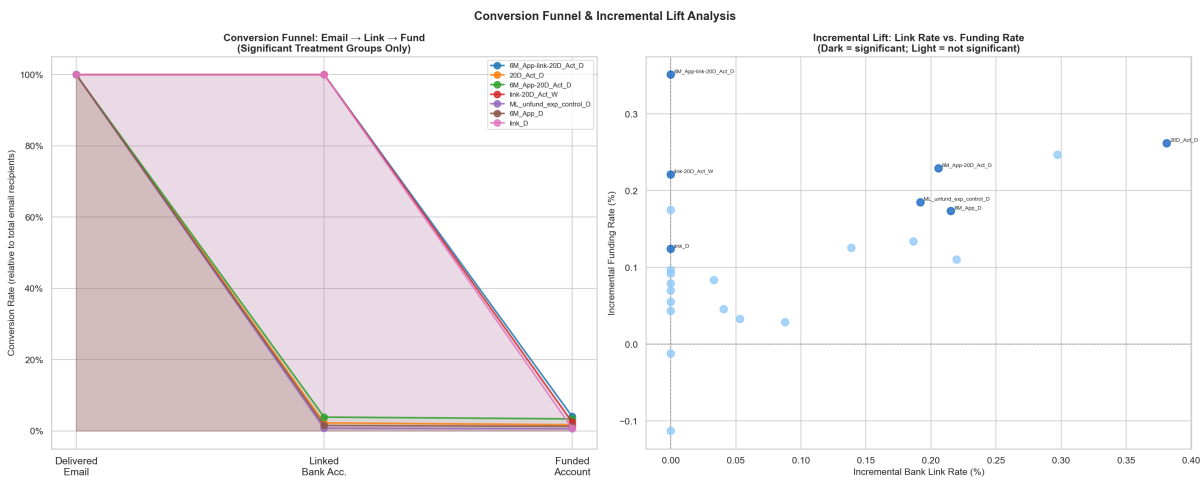

Figure 9: Left: Conversion funnel curves for statistically significant treatment groups. Right: Incremental lift scatter plot — groups in the upper-right quadrant show the highest email-attributable gains in both link and funding rates.

10.2 Funnel Insights

Insight 1 — Bank linkage is a leading indicator for funding. Across all groups, users who link their bank account convert to funding at much higher rates than those who don't. The `link_flag` segment, where all users have already linked accounts, shows a clear advantage — the largest conversion bottleneck for these users is awareness and motivation, not operational friction.

Insight 2 — Email-attributable link uplift correlates with funding uplift. The scatter plot (Figure 9, right) shows a positive relationship between incremental link rate and incremental funding rate across significant groups. This validates the causal chain: emails → higher bank linkage → higher funding.

Insight 3 — Highest-baseline groups are over-represented in non-significant results. Groups like `6M_App-link-20D_Act-5D_Act` have extremely high baseline funding rates but do not show significant email lift. These users are already highly motivated — the email provides no meaningful incremental nudge. This argues for **segment-specific investment allocation**: email budget should flow toward medium-intent segments where activation energy is achievable, not high-intent segments who convert organically.

11 Business Recommendations & Next Steps

11.1 Immediate Actions (Post-Experiment)

1. **Scale daily delivery to significant treatment groups.** The following segments are confirmed to respond positively to daily email campaigns and should be included in the next production email run: `20D_Act_D`, `6M_App_D`, `6M_App-20D_Act_D`, `6M_App-link-20D_Act_D`, `ML_unfund_exp_control_D`, `link_D`.
2. **Prioritize `ml_funding_faq` as the anchor template.** Given its dominant open rate (∼31.5%), this template should be placed at **position 1** in the delivery sequence (leveraging the priming effect) and potentially personalized further with segment-specific FAQ content.
3. **Reduce email exposure for high-baseline, non-significant segments.** Segments such as `6M_App-link-20D_Act-5D_Act` convert organically. Email exposure here creates unsubscribe risk without incremental benefit. Redirect these users to in-app notifications or push campaigns.

11.2 Near-Term Optimization (30-60 Days)

4. **A/B test subject line variations for `ml_funding_faq`.** The template's content clearly resonates, but the open rate may be further improved by optimizing the subject line for different segments. Recommended subject line hypothesis: segment-specific questions (e.g., "How do you fund your [Segment Name] account? 3 questions answered").
5. **Explore optimal send time.** All emails in this experiment were sent between 10:00–10:30 AM. A/B testing evening send windows (6:00–8:00 PM) may improve open rates for users who check email during commute or evening relaxation time.
6. **Implement re-engagement trigger for openers who didn't fund.** Users who opened 2+ emails but did not fund within 72 hours represent a "high-intent, unresolved" cohort. A triggered push notification or in-app message at the 72-hour mark may convert this population efficiently.

11.3 Long-Term Strategic Direction

7. **Build a personalized email sequencing model.** The current experiment uses randomized email ordering to explore the full combinatorial space. The next step is training an ML model (e.g., multi-armed bandit or contextual bandits) to dynamically select the next-best email for each user based on their engagement history, segment features, and predicted conversion probability.
8. **Extend to multi-channel orchestration.** Email is one channel in a broader engagement stack. Coordinating email campaigns with push notifications, in-app banners, and SMS (for high-intent moments) can create a multiplicative engagement effect. The segment-level insights from this experiment provide a rigorous foundation for cross-channel targeting logic.

12 Conclusion

This experiment provides rigorous, statistically-grounded evidence that targeted email campaigns can causally improve funding rates for approved-but-unfunded fintech users. Key conclusions:

- **7 of 24 treatment arms achieve statistical significance** ($p \leq 0.05$), generating an estimated **500+** email-attributable funded accounts.
- **Daily delivery dominates twice-a-week delivery** for unfunded user activation. Frequency is a significant lever independent of content quality.

- **FAQ-style content that directly addresses funding concerns** (the `ml_funding_faq` template) outperforms motivational, educational, and app-exploration content by a substantial margin — consistent with the hypothesis that knowledge barriers are the primary obstacle.
- **The second email in sequence (order_1) outperforms the first (order_0)**, invalidating the novelty hypothesis and suggesting a priming effect. Email sequences should be designed to capitalize on this by placing key content at position 1 rather than position 0.
- **High-intent segments exhibit diminishing email returns** and should be targeted via more frictionless channels (in-app, push) to avoid deliverability erosion.
- **The experiment is not yet complete for weekly groups** — final conclusions for twice-a-week delivery arms must await the Day_32 email delivery and the post-experiment observation window.

The analytical framework established in this project — stratified segmentation, multi-arm A/B testing with matched controls, causal attribution via proportions z-tests, and time-series engagement analysis — provides a replicable template for future growth experiments at scale.

12 Appendix: Technical Implementation Notes

12.0 A. Statistical Test Selection

We use a **one-sided proportions Z-test** (rather than a two-sided test) because our directional hypothesis is specifically that email intervention *increases* funding rate. Using a one-sided test is appropriate here since:

1. The business question is directional: we seek to confirm positive impact.
2. A two-sided test would sacrifice statistical power for detecting increases.
3. The control comparison provides a clean counterfactual (not a pre/post design).

The test is implemented via `statsmodels.stats.proportion.proportions_ztest` with `alternative='larger'`.

12.0 B. Control Group Matching

Weekly (`_W`) treatment groups are matched to the daily (`_D`) control baseline of the same segment. This is because control groups are maintained at the segment level (not frequency level) — the control users simply do not receive any emails regardless of what frequency

they would have been assigned to.

12.0 C. Open Rate Computation Caveat

Open rate is computed from the *summary status* in `user_events.csv`, where each user carries the highest-intent action status for each email. This means "delivered" includes all users who received the email but did not open it. The resulting open rates are user-level open rates (not impression-level), which is the correct metric for behavioral analysis.

12.0 D. Time Series Vectorized Computation

The time series analysis requires a "gather" operation: for each user at delivery position d, look up the email status for whichever template was assigned at that position. This is implemented via a vectorized mask loop over the 10 email templates to avoid the performance penalty of row-wise `apply`.

12 References

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